PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Software Requirements Specification

for

Online Internship & Placement Preparation Portal (OIPPP)

Version 1.0 approved

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PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

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Revision History

Name	Date	Reason For Changes	Version
Sharath H G	18-09-2026	First draft of sections 1 and 2	0.1
All members	26-09-2026	Feature sections and NFRs added by each owner	0.2
Guru R Bharadwaj	01-10-2026	Traceability matrix, review fixes	0.3
All members	02-10-2026	Approved for submission	1.0

1. Introduction

1.1 Purpose

This document specifies the software requirements for release 1.0 of the Online Internship & Placement Preparation Portal (OIPPP). It covers the whole system: the web client, the API server, the database and the small analytics service. It is the basis for the architecture and design document, the test plan and the sprint backlog.

1.2 Intended Audience

The readers are the four developers on the team, the course faculty who evaluate the project, testers who will write test cases from it, and the college placement cell, which is the intended customer. Section 2 gives an overview and is enough for faculty and the placement cell. Sections 3 to 5 are written for developers and testers. Appendix C links every requirement to design, code and tests.

1.3 Product Scope

At PES today, internship openings reach students through notice boards, WhatsApp groups, mail threads and spreadsheets kept by the placement office. Aptitude practice happens on outside websites. The placement office has no single place to see who applied where or how ready a student is.

OIPPP brings listings, applications and aptitude practice behind one login. Students search listings, apply and track each application, and take timed practice tests. Recruiters post listings and move applicants through their stages. The placement admin approves recruiters, writes tests and gets batch-wide placement numbers. The goal is to cut the time students spend chasing information and to give the placement cell data it currently collects by hand.

Out of scope for 1.0: video interviews, payment of any kind, a mobile app, and integration with the university ERP.

1.4 References

- PES University, SRS-Document template, Dept. of CSE, 2026.
- PES University, Project Guidelines (mini project proposal handout), 2026.
- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications.
- OWASP Top 10 (2021), https://owasp.org/Top10/

- OIPPP Project Proposal and CONTRIBUTING.md, team GitHub repository.
- React (https://react.dev), Express (https://expressjs.com), MongoDB (https://www.mongodb.com/docs) documentation.

2. Overall Description

2.1 Product Perspective

OIPPP is a new, self-contained product. It does not replace an existing system, though it takes over work the placement office does with spreadsheets and email. It is a three-tier web application: a React single-page client, a Node.js/Express REST API, and MongoDB. A small Python service computes readiness scores and batch statistics. The only outside system is an SMTP mail provider. Figure 1 shows the major components.

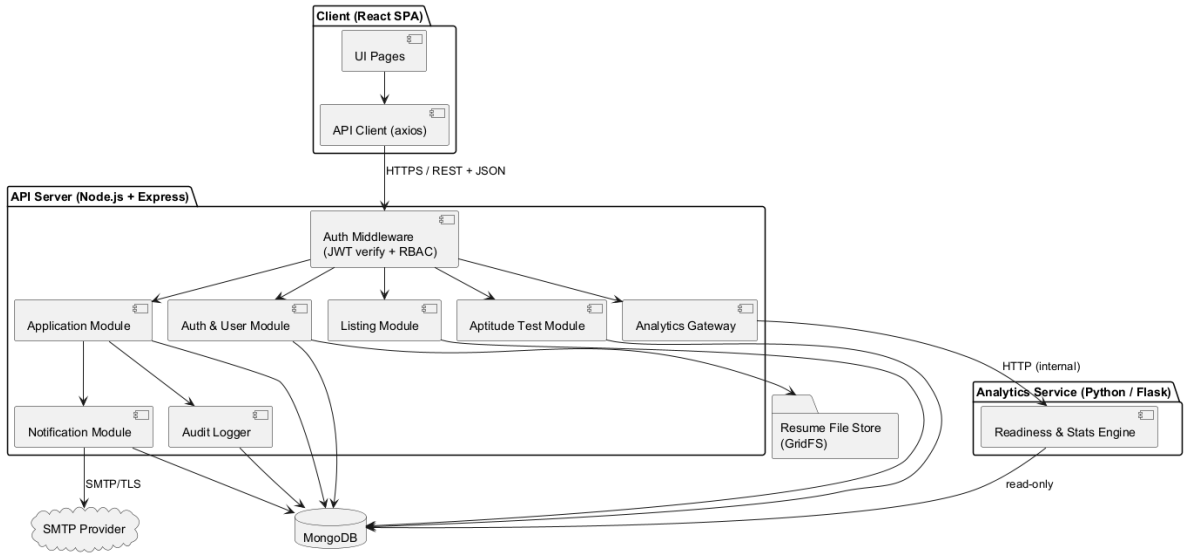


Figure 1: Major components and external interfaces

2.2 Product Functions

Identity and administration

- Register and log in as Student or Recruiter; admins are created by seed script.
- Student profile with academic details, skills and one resume.
- Admin approves recruiters and disables accounts.

Listings and discovery

- Recruiters post, edit and close internship listings; listings close on their deadline.
- Students search, filter, sort and bookmark listings and get a skill-matched home feed.

Applications

- Students apply once per listing and follow status from Applied to Offer or Rejected.
- Recruiters review applicants and change status; students are notified in-app and by email.

Preparation and analytics

- Admins build aptitude tests from a question bank.
- Students take timed tests with auto-save and see per-topic results.
- Students see their progress; the admin sees placement statistics and exports a CSV report.

2.3 User Classes and Characteristics

User class	Who	Frequency	Technical skill	Privilege
Student (most important)	UG/PG students in their placement year, about 2,000 per batch	Daily during the season	Comfortable with web apps	Own profile, applications and attempts
Recruiter	HR staff or engineers from visiting companies	Weekly, heavier near deadlines	Varies, often low	Own listings and their applicants
Placement Admin	Placement office staff, 2 to 4 people	Daily	Moderate	Full access, approvals, tests, statistics

2.4 Operating Environment

- Client: latest two versions of Chrome, Firefox and Edge on Windows, macOS, Linux and Android; screens 360 px wide and up.
- Server: Node.js 20 LTS on Ubuntu 22.04 (Render free tier for demo); MongoDB 7 (Atlas M0 cluster).
- Analytics: Python 3.11 with Flask 3, deployed next to the API.
- Development: Windows 11 and Ubuntu laptops, VS Code, Git and GitHub, Jira for sprint tracking.

2.5 Design and Implementation Constraints

- The problem statement fixes the stack: MongoDB, Express, React, Node.js, with Python allowed for analytics.
- Only free services can be used (Atlas M0 has 512 MB storage, Render free instances sleep when idle).
- Communication between client and server is REST over HTTPS with JSON bodies.
- The project must be finished within the semester by a team of four, in two-week sprints.
- Code follows the ESLint recommended rules and the commit and branch conventions in CONTRIBUTING.md.

2.6 Assumptions and Dependencies

- Students have a stable internet connection during a timed test. SAFE-1 limits the damage when they do not.
- Recruiters post genuine listings; the admin approval step is the only check.
- The free SMTP provider (Brevo, 300 mails/day) is enough for the demo. If not, email falls back to in-app only.
- The college gives each student a unique SRN and an institutional email address.

3. External Interface Requirements

3.1 User Interfaces

Every page has the same top bar with the logo, role-specific menu, notification bell and profile menu, and a footer with a Help link. Errors from the server appear as a red banner at the top of the form with a plain message, and field errors appear under the field. Main screens:

Screen	Role	Contents
Login / Register	All	Email, password, role choice, SRN for students

Screen	Role	Contents
Home feed	Student	Top 10 recommended listings, search bar, deadlines this week
Listing search and detail	Student	Filters panel, result cards, bookmark icon, Apply button
My Applications	Student	Table of applications with status chips and history drawer
Test list and attempt	Student	Available tests, timer, question palette, auto-save indicator
Progress	Student	Score trend chart, weak topics, application counts
My Listings and Applicants	Recruiter	Listing form, applicant table, resume viewer, status actions
Admin console	Admin	Pending recruiters, users, question bank, test builder, statistics

3.2 Software Interfaces

Component	Version	Data in / out	Purpose
MongoDB via Mongoose	MongoDB 7, Mongoose 8	All persistent documents	Storage of users, listings, applications, tests, attempts, notifications, audit log
GridFS	MongoDB 7	Resume PDF files	File storage without a separate service
jsonwebtoken / bcrypt	9.x / 5.x	Tokens, password hashes	Authentication
Nodemailer + SMTP provider	6.x	Out: recipient, subject, body	Status change emails
Analytics service (Flask)	Python 3.11, Flask 3	In: studentId or filter; out: readiness and statistics JSON	Readiness score and batch statistics

3.3 Communications Interfaces

The browser talks to the API with HTTPS (TLS 1.2+) and JSON. The token travels in the Authorization header as a Bearer token. The API calls the analytics service over HTTP on the private network and sends email over SMTP with STARTTLS. Resume downloads are streamed with Content-Type application/pdf. Request bodies are limited to 100 KB, except uploads, which are limited to 2 MB.

3.4 Analysis Models

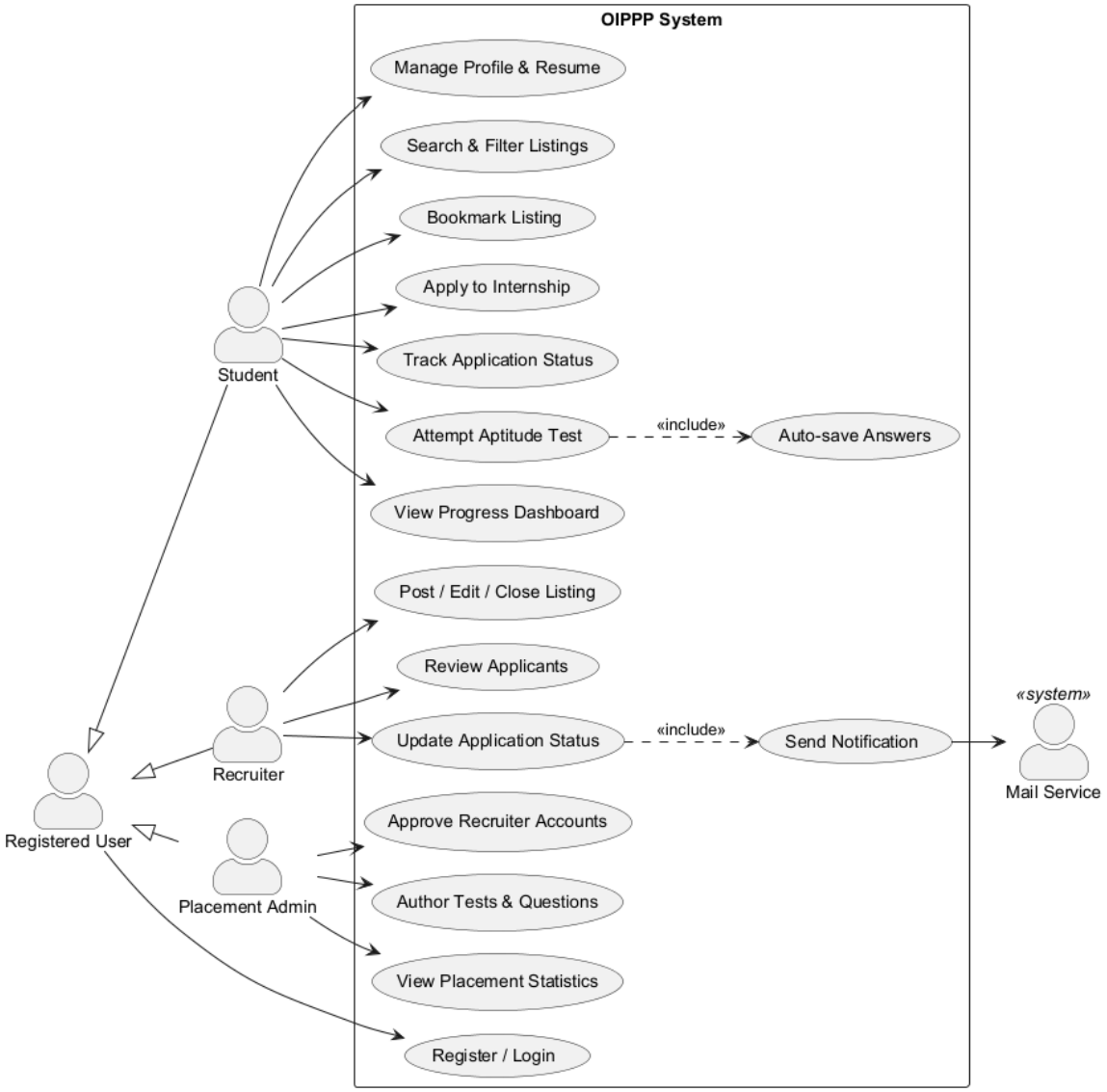


Figure 2: Use case diagram

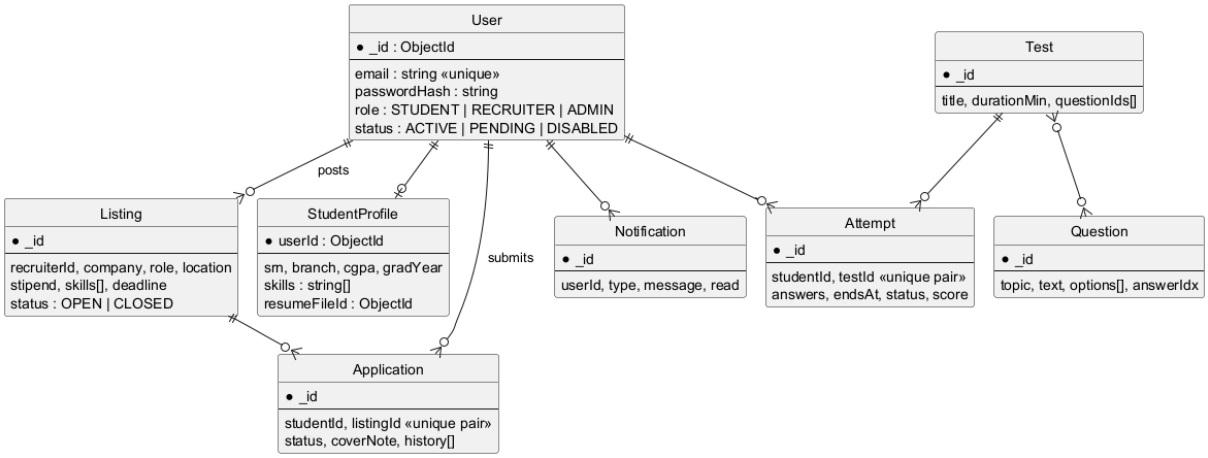


Figure 3: Entity-relationship model of the main collections

4. System Features

4.1 Authentication and Role-Based Access

4.1.1 Description and Priority

Lets students, recruiters and placement admins create an account and sign in. Every protected page and API route checks the caller's role before doing anything. Without this nothing else in the portal can be trusted, so it is built first. Priority: High (benefit 9, penalty 9, cost 4, risk 5). Owner: Deekshith Gowda G.

4.1.2 Stimulus/Response Sequences

- User fills the registration form and submits. The system validates the fields, stores the account (students become ACTIVE, recruiters become PENDING) and shows a confirmation.
- User enters email and password on the login page. On success the system issues a session token and opens the dashboard for that role. On failure it shows "Invalid email or password" without saying which one was wrong.
- User opens a page that belongs to another role. The system returns 403 and the client shows the "You do not have access to this page" screen.

4.1.3 Functional Requirements

AUTH-1: The system shall allow a new user to register as Student or Recruiter using name, email and password. A Student must also give a valid SRN of the form PES[1-2]UG[0-9]{2}[A-Z]{2}[0-9]{3}.

AUTH-2: The system shall reject a registration whose email is already in use and display "An account with this email already exists".

AUTH-3: The system shall reject passwords shorter than 8 characters or lacking at least one letter and one digit, and show the rule that failed.

AUTH-4: The system shall authenticate a user by email and password and issue a signed JWT that expires 60 minutes after issue.

AUTH-5: The system shall lock an account for 15 minutes after 5 consecutive failed login attempts and tell the user when they can retry.

AUTH-6: The system shall check the role claim of the token on every protected route and return HTTP 403 when the role is not permitted for that route.

4.2 Profile and User Administration

4.2.1 Description and Priority

Students keep an academic profile and one current resume. Placement admins approve recruiter accounts and can disable any account. Priority: High (benefit 8, penalty 7, cost 4, risk 3). Owner: Deekshith Gowda G.

4.2.2 Stimulus/Response Sequences

- Student edits profile fields and uploads a resume. The system validates the file and saves it, replacing any older resume.
- Admin opens the pending recruiter list and approves or rejects one. The recruiter receives an email with the decision.
- Admin disables an account. That user's next request is refused and they are signed out.

4.2.3 Functional Requirements

PROF-1: The system shall let a student view and edit branch, CGPA (0.00 to 10.00), graduation year and a skill list of up to 20 skills.

PROF-2: The system shall accept resume uploads only in PDF format up to 2 MB and reject anything else with the reason shown.

PROF-3: The system shall let an admin approve or reject a PENDING recruiter account, recording a reason code when rejecting.

PROF-4: The system shall let an admin disable an account, after which all tokens issued to that account are refused.

4.3 Internship Listing Management

4.3.1 Description and Priority

Approved recruiters publish internship listings, edit them while open, and close them. A listing closes itself when its deadline passes. Priority: High (benefit 9, penalty 8, cost 3, risk 3). Owner: Sharath H G.

4.3.2 Stimulus/Response Sequences

- Recruiter fills the listing form and publishes. The listing appears in student search within one minute.
- Recruiter edits an open listing. Students who bookmarked it see the updated details.
- Deadline passes. The system marks the listing CLOSED and stops accepting applications.

4.3.3 Functional Requirements

LST-1: The system shall let an approved recruiter create a listing with company, role title, location, mode (onsite/remote/hybrid), stipend, required skills, eligibility CGPA and application deadline.

LST-2: The system shall reject a listing whose deadline is in the past or whose stipend is negative.

LST-3: The system shall allow only the recruiter who created a listing to edit or close it.

LST-4: The system shall change a listing's status to CLOSED automatically once its deadline has passed, within 5 minutes of the deadline.

4.4 Search, Filtering, Bookmarks and Recommendations

4.4.1 Description and Priority

Students find listings by keyword and filters, save the ones they care about, and get a feed ranked by how well each listing matches their skills. Priority: High (benefit 8, penalty 6, cost 5, risk 4). Owner: Sharath H G.

4.4.2 Stimulus/Response Sequences

- Student types a keyword and selects filters. The system returns matching OPEN listings, 20 per page.
- Student clicks the bookmark icon. The listing is added to Saved Internships.
- Student opens the home page. The system shows the top 10 listings ranked by skill match.

4.4.3 Functional Requirements

SRCH-1: The system shall search OPEN listings by keyword across role title, company and skills.

SRCH-2: The system shall filter listings by location, mode, minimum stipend, skill and deadline range, and sort by deadline, stipend or posting date.

SRCH-3: The system shall return search results in pages of 20 and show the total count.

SRCH-4: The system shall let a student bookmark and remove a bookmark on a listing, with at most 100 bookmarks per student.

SRCH-5: The system shall rank the home feed by the share of a listing's required skills present in the student's profile and hide listings whose eligibility CGPA exceeds the student's CGPA.

4.5 Application Submission and Tracking

4.5.1 Description and Priority

A student applies to an open listing with their resume and a cover note and then follows the application through its stages. Priority: High (benefit 9, penalty 9, cost 5, risk 4). Owner: Guru R Bharadwaj.

4.5.2 Stimulus/Response Sequences

- Student opens an open listing and presses Apply. The system checks the deadline, eligibility and earlier applications, stores the application as APPLIED and notifies the recruiter.
- Student presses Apply on a listing they already applied to. The system refuses and shows the existing application.
- Student opens My Applications. The system lists every application with its current status and full history.

4.5.3 Functional Requirements

APP-1: The system shall let an authenticated student submit exactly one application per listing, attaching the current resume and an optional cover note of up to 1000 characters.

APP-2: The system shall reject an application to a listing that is CLOSED or past its deadline and display "Applications for this internship are closed".

APP-3: The system shall reject an application from a student whose CGPA is below the listing's eligibility CGPA and display the required value.

APP-4: The system shall record every status change of an application with the old status, new status, actor and timestamp.

APP-5: The system shall show a student all of their applications with current status and history, newest first.

4.6 Recruiter Review and Notifications

4.6.1 Description and Priority

Recruiters review applicants for their own listings and move them through the stages. Each change reaches the student in the app and by email. Priority: Medium (benefit 7, penalty 7, cost 4, risk 4). Owner: Guru R Bharadwaj.

4.6.2 Stimulus/Response Sequences

- Recruiter opens a listing's applicant list, views a resume and shortlists or rejects the applicant.
- Recruiter rejects an applicant. The system asks for a reason before saving.
- Any status change. The student sees an in-app notification immediately and gets an email.

4.6.3 Functional Requirements

REV-1: The system shall show a recruiter the applicants of their own listings only, with profile summary and resume link.

REV-2: The system shall allow only these transitions: APPLIED to SHORTLISTED or REJECTED; SHORTLISTED to INTERVIEW or REJECTED; INTERVIEW to OFFER or REJECTED. Any other transition is refused with HTTP 422.

REV-3: The system shall require a rejection reason chosen from a fixed list when an application is moved to REJECTED.

REV-4: The system shall create an in-app notification and send an email to the student within 2 minutes of every status change, retrying a failed email up to 3 times.

4.7 Aptitude Test Engine

4.7.1 Description and Priority

Admins build tests from a question bank. Students take each test once, under a timer, and see a scored result broken down by topic. Priority: High (benefit 8, penalty 7, cost 6, risk 6). Owner: E S Sukruth.

4.7.2 Stimulus/Response Sequences

- Admin adds questions with topic and correct option, then assembles a test with a duration.
- Student starts a test. The timer starts on the server and every answer is saved as it is picked.
- Student submits, or time runs out. The system scores the attempt and shows the result per topic.

4.7.3 Functional Requirements

TST-1: The system shall let an admin create multiple-choice questions with 2 to 6 options, exactly one correct option, a topic (Quantitative, Logical, Verbal or Technical) and a difficulty level.

TST-2: The system shall let an admin assemble a test from 5 to 60 questions with a duration between 10 and 120 minutes.

TST-3: The system shall allow a student exactly one submitted attempt per test and return the in-progress attempt if the student starts the same test again.

TST-4: The system shall save each answer to the server within 1 second of selection while an attempt is in progress.

TST-5: The system shall auto-submit an attempt when its server-side end time passes and reject answers received more than 30 seconds after that time.

TST-6: The system shall score an attempt immediately on submission and show total score and per-topic percentage.

4.8 Progress Tracking and Analytics

4.8.1 Description and Priority

Students see how they are doing over time. The placement admin sees placement numbers across the batch. Priority: Medium (benefit 7, penalty 5, cost 6, risk 5). Owner: E S Sukruth.

4.8.2 Stimulus/Response Sequences

- Student opens the Progress page. The system shows test scores over time, topic strengths and application outcomes.
- Admin opens the Placement Statistics page and picks a filter. The system shows counts and rates for that slice.

4.8.3 Functional Requirements

ANL-1: The system shall show each student a readiness view with test score trend, weakest two topics and counts of applications per status.

ANL-2: The system shall show the admin placement statistics: students with at least one offer, offers per company, applications per listing and average test score per branch, filterable by branch and graduation year.

ANL-3: The system shall let the admin export the placement report as CSV.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

Students check listings in bursts right after a company announces a drive, so search speed matters most.

PERF-1: Filtered listing search shall return the first page in under 2 seconds at the 95th percentile with 5,000 listings in the database and 50 concurrent users.

PERF-2: Any other API call except file upload shall respond in under 1 second at the 95th percentile under the same load.

PERF-3: The analytics dashboards shall render in under 3 seconds for a batch of 2,000 students.

5.2 Safety Requirements

No physical harm is possible. The losses that matter are a student losing a test attempt and the college losing placement records.

SAFE-1: No answer saved during a test attempt shall be lost if the browser is refreshed, closed or disconnected; the attempt resumes with saved answers and the remaining time.

SAFE-2: Disabling an account shall not delete its applications or attempts, so placement records stay complete.

5.3 Security Requirements

Security objectives

SO-1: Confidentiality: a user can read only the data their role and ownership allow. Resumes in particular are visible only to the owner, recruiters of listings the student applied to, and admins.

SO-2: Integrity: application statuses and test scores can be changed only by the actors and through the transitions the business rules permit, and every change is traceable.

SO-3: Authentication: every request that reads or changes data comes from a verified, active account.

Security requirements

SEC-1: Passwords shall be stored only as bcrypt hashes with cost factor 10 or higher; plain passwords shall never be stored or logged.

SEC-2: All client-server traffic shall use HTTPS with TLS 1.2 or higher.

SEC-3: A resume file shall be served only to its owner, to recruiters of listings that student applied to, and to admins; all other requests get HTTP 403.

SEC-4: All request bodies shall be validated against a schema, and query operators supplied by the client (for example keys starting with $) shall be rejected to prevent NoSQL injection.

SEC-5: Login and registration endpoints shall be rate limited to 10 requests per minute per IP address.

SEC-6: Every application status change and every admin action on accounts shall be written to an audit log that cannot be edited through the API.

Personal data (name, SRN, CGPA, resume) is handled according to the Digital Personal Data Protection Act, 2023: it is collected only for placement use and a student can ask the admin to delete their account after the season.

5.4 Software Quality Attributes

QA-1: Usability: a first-time student shall be able to find a listing and apply within 3 minutes without help, checked with 5 classmates during acceptance testing.

QA-2: Availability: the portal shall be available 99% of the time during the placement season, excluding announced maintenance.

QA-3: Reliability: no application shall be lost or duplicated under concurrent submissions (enforced by a unique index on student and listing).

QA-4: Maintainability: server code shall have at least 70% statement coverage from automated tests, and each module shall keep its routes, service and model in its own folder.

QA-5: Portability: the client shall work on the latest two versions of Chrome, Firefox and Edge and on screens 360 px wide and up.

Where these conflict, reliability of applications and test attempts comes before speed, and ease of use comes before feature count.

5.5 Business Rules and Domain Requirements

Business rules

- BR-1: Only recruiters approved by a placement admin can post listings.
- BR-2: Only the recruiter who owns a listing can change the status of its applications.
- BR-3: A student may apply to a listing only once and attempt a given test only once.
- BR-4: Only admins create questions and tests and view batch-wide statistics.
- BR-5: A student who has accepted an offer can still apply to other listings unless the admin turns on the one-offer rule (see DOM-2).

Domain requirements

- DOM-1: Eligibility is based on CGPA; a listing may set a minimum CGPA and the system enforces it at apply time.
- DOM-2: The college may enforce a one-offer policy. When the admin enables it, a student holding an OFFER cannot submit new applications.
- DOM-3: The placement season runs per graduation year; statistics are always reported per graduation year.

6. Other Requirements

- Database: every collection has a schema in Mongoose; unique indexes on user email, (studentId, listingId) in applications and (studentId, testId) in attempts; text index on listing title, company and skills.
- Backups: Atlas daily snapshot, kept 7 days.
- Internationalization: English only for 1.0; dates shown as DD-MM-YYYY and money in INR.
- Reuse: no code is copied from other projects; open-source libraries are listed in package.json with their licences (all MIT or Apache 2.0).
- Legal: the project is released under the MIT licence.

Appendix A: Glossary

Term	Meaning
Listing	An internship opening posted by a recruiter
Application	A student's request to be considered for one listing
Attempt	One student's sitting of one aptitude test

Term	Meaning
Readiness	Score from 0 to 100 combining test results and application outcomes
SRN	Student Registration Number issued by PES University
JWT	JSON Web Token, the signed session token
RBAC	Role-based access control
CGPA	Cumulative grade point average, 0 to 10
GridFS	MongoDB's way of storing files larger than a document
NFR	Non-functional requirement
SPA	Single-page application
TLS	Transport Layer Security

Appendix B: Field Layouts

Student registration

Field	Length	Data Type	Description	Is Mandatory
SRN	13	Alphanumeric	PES registration number	Y
Full Name	60	String		Y
Email	60	String	Institutional email, unique	Y
Password	8-64	String	Stored as bcrypt hash	Y
Branch	30	String	e.g. CSE, ECE	Y
CGPA	4	Numeric	0.00 to 10.00	Y
Graduation Year	4	Numeric		Y
Skills	20 x 30	String list		N
Resume File ID	24	Alphanumeric	GridFS id of current resume	N
Account Status	25	Alphanumeric	ACTIVE / PENDING / DISABLED	Y
Reject Reason Code	4	String	Set when a recruiter account is rejected	N

Internship listing

Field	Length	Data Type	Description	Is Mandatory
Company	80	String		Y
Role Title	80	String		Y
Location	60	String		Y
Mode	8	Enum	ONSITE / REMOTE / HYBRID	Y
Stipend	7	Numeric	INR per month, 0 if unpaid	Y
Required Skills	10 x 30	String list		Y
Eligibility CGPA	4	Numeric		N
Deadline	8	Date	Last date to apply	Y

Field	Length	Data Type	Description	Is Mandatory
Status	6	Enum	OPEN / CLOSED	Y

Report requirements

Placement Report	Application Report
SRN	Application ID
Student Name	SRN
Branch	Student Name
Graduation Year	Company
CGPA	Role Title
Applications Count	Applied Date
Offers Count	Current Status
Companies with Offer	Last Status Change Date
Average Test Score	Rejection Reason
Readiness Score	Recruiter

Appendix C: Requirement Traceability Matrix

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
1	AUTH-1	Allow a new user to register as Student or Recruiter using name,...	Auth & User Module, Auth Middleware	SAD 4.3.1 Auth API	server/src/modules/auth/	TC-AUTH-01	ST-AUTH-1
2	AUTH-2	Reject a registration whose email is already in use and display...	Auth & User Module, Auth Middleware	SAD 4.3.1 Auth API	server/src/modules/auth/	TC-AUTH-01	ST-AUTH-2
3	AUTH-3	Reject passwords shorter than 8 characters or lacking at least one...	Auth & User Module, Auth Middleware	SAD 4.3.1 Auth API	server/src/modules/auth/	TC-AUTH-02	ST-AUTH-3
4	AUTH-4	Authenticate a user by email and password and issue a signed JWT...	Auth & User Module, Auth Middleware	SAD 4.3.1 Auth API	server/src/modules/auth/	TC-AUTH-03	ST-AUTH-4

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
5	AUTH-5	Lock an account for 15 minutes after 5 consecutive failed login...	Auth & User Module, Auth Middleware	SAD 4.3.1 Auth API	server/src/modules/auth/	TC-AUTH-03	ST-AUTH-5
6	AUTH-6	Check the role claim of the token on every protected route and...	Auth & User Module, Auth Middleware	SAD 4.3.1 Auth API	server/src/modules/auth/	TC-AUTH-04	ST-AUTH-6
7	PROF-1	Let a student view and edit branch, CGPA (0.00 to 10.00),...	Auth & User Module, Resume File Store	SAD 4.3.1 Auth API	server/src/modules/users/	Unit tests	ST-PROF-1
8	PROF-2	Accept resume uploads only in PDF format up to 2 MB and reject...	Auth & User Module, Resume File Store	SAD 4.3.1 Auth API	server/src/modules/users/	TC-PROF-01	ST-PROF-2
9	PROF-3	Let an admin approve or reject a PENDING recruiter account,...	Auth & User Module, Resume File Store	SAD 4.3.1 Auth API	server/src/modules/users/	Unit tests	ST-PROF-3
10	PROF-4	Let an admin disable an account, after which all tokens issued to...	Auth & User Module, Resume File Store	SAD 4.3.1 Auth API	server/src/modules/users/	Unit tests	ST-PROF-4
11	LST-1	Let an approved recruiter create a listing with company, role...	Listing Module	SAD 4.3.2 Listings API	server/src/modules/listings/	TC-LST-01	ST-LST-1
12	LST-2	Reject a listing whose deadline is in the past or whose stipend is...	Listing Module	SAD 4.3.2 Listings API	server/src/modules/listings/	TC-LST-01	ST-LST-2

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
13	LST-3	Allow only the recruiter who created a listing to edit or close it.	Listing Module	SAD 4.3.2 Listings API	server/src/modules/listings/	Unit tests	ST-LST-3
14	LST-4	Change a listing's status to CLOSED automatically once its...	Listing Module	SAD 4.3.2 Listings API	server/src/modules/listings/	TC-LST-01	ST-LST-4
15	SRCH-1	Search OPEN listings by keyword across role title, company and...	Listing Module	SAD 4.3.2 Listings API	server/src/modules/listings/	TC-SRCH-01	ST-SRCH-1
16	SRCH-2	Filter listings by location, mode, minimum stipend, skill and...	Listing Module	SAD 4.3.2 Listings API	server/src/modules/listings/	TC-SRCH-01	ST-SRCH-2
17	SRCH-3	Return search results in pages of 20 and show the total count.	Listing Module	SAD 4.3.2 Listings API	server/src/modules/listings/	TC-SRCH-01	ST-SRCH-3
18	SRCH-4	Let a student bookmark and remove a bookmark on a listing, with at...	Listing Module	SAD 4.3.2 Listings API	server/src/modules/listings/	Unit tests	ST-SRCH-4
19	SRCH-5	Rank the home feed by the share of a listing's required skills...	Listing Module	SAD 4.3.2 Listings API	server/src/modules/listings/	Unit tests	ST-SRCH-5
20	APP-1	Let an authenticated student submit exactly one application per...	Application Module, Audit Logger	SAD 4.2.1 seq. diagram, 4.3.3	server/src/modules/applications/	TC-APP-01	ST-APP-1

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
21	APP-2	Reject an application to a listing that is CLOSED or past its...	Application Module, Audit Logger	SAD 4.2.1 seq. diagram, 4.3.3	server/src/modules/applications/	TC-APP-02	ST-APP-2
22	APP-3	Reject an application from a student whose CGPA is below the...	Application Module, Audit Logger	SAD 4.2.1 seq. diagram, 4.3.3	server/src/modules/applications/	TC-APP-02	ST-APP-3
23	APP-4	Record every status change of an application with the old status,...	Application Module, Audit Logger	SAD 4.2.1 seq. diagram, 4.3.3	server/src/modules/applications/	TC-APP-01	ST-APP-4
24	APP-5	Show a student all of their applications with current status and...	Application Module, Audit Logger	SAD 4.2.1 seq. diagram, 4.3.3	server/src/modules/applications/	Unit tests	ST-APP-5
25	REV-1	Show a recruiter the applicants of their own listings only, with...	Application Module, Notification Module	SAD 4.2.3 seq. diagram, 4.3.3	server/src/modules/applications/	Unit tests	ST-REV-1
26	REV-2	Allow only these transitions: APPLIED to SHORTLISTED or REJECTED;...	Application Module, Notification Module	SAD 4.2.3 seq. diagram, 4.3.3	server/src/modules/applications/	TC-REV-01	ST-REV-2
27	REV-3	Require a rejection reason chosen from a fixed list when an...	Application Module, Notification Module	SAD 4.2.3 seq. diagram, 4.3.3	server/src/modules/applications/	TC-REV-01	ST-REV-3
28	REV-4	Create an in-app notification and send an email to the student...	Application Module, Notification Module	SAD 4.2.3 seq. diagram, 4.3.3	server/src/modules/applications/	TC-REV-01	ST-REV-4

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
29	TST-1	Let an admin create multiple-choice questions with 2 to 6 options,...	Aptitude Test Module	SAD 4.2.2 seq. diagram, 4.3.4	server/src/modules/tests/	Unit tests	ST-TST-1
30	TST-2	Let an admin assemble a test from 5 to 60 questions with a...	Aptitude Test Module	SAD 4.2.2 seq. diagram, 4.3.4	server/src/modules/tests/	Unit tests	ST-TST-2
31	TST-3	Allow a student exactly one submitted attempt per test and return...	Aptitude Test Module	SAD 4.2.2 seq. diagram, 4.3.4	server/src/modules/tests/	TC-TST-01	ST-TST-3
32	TST-4	Save each answer to the server within 1 second of selection while...	Aptitude Test Module	SAD 4.2.2 seq. diagram, 4.3.4	server/src/modules/tests/	TC-TST-01	ST-TST-4
33	TST-5	Auto-submit an attempt when its server-side end time passes and...	Aptitude Test Module	SAD 4.2.2 seq. diagram, 4.3.4	server/src/modules/tests/	TC-TST-02	ST-TST-5
34	TST-6	Score an attempt immediately on submission and show total score...	Aptitude Test Module	SAD 4.2.2 seq. diagram, 4.3.4	server/src/modules/tests/	TC-TST-02	ST-TST-6
35	ANL-1	Show each student a readiness view with test score trend, weakest...	Analytics Gateway, Analytics Service	SAD 4.3.5	analytics/app.py	Unit tests	ST-ANL-1
36	ANL-2	Show the admin placement statistics: students with at least one...	Analytics Gateway, Analytics Service	SAD 4.3.5	analytics/app.py	TC-ANL-01	ST-ANL-2

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37	ANL-3	Let the admin export the placement report as CSV.	Analytics Gateway, Analytics Service	SAD 4.3.5	analytics/app.py	TC-ANL-01	ST-ANL-3
38	PERF-1	Filtered listing search shall return the first page in under 2...	SAD 3.9 / 3.7	SAD 4.4	server/src/middleware/	TC-PERF-01	ST-PERF-1
39	PERF-2	Any other API call except file upload shall respond in under 1...	SAD 3.9 / 3.7	SAD 4.4	server/src/middleware/	TC-PERF-01	ST-PERF-2
40	SAFE-1	No answer saved during a test attempt shall be lost if the browser...	SAD 3.9 / 3.7	SAD 4.4	server/src/middleware/	TC-TST-01	ST-SAFE-1
41	SEC-1	Passwords shall be stored only as bcrypt hashes with cost factor...	SAD 3.9 / 3.7	SAD 4.4	server/src/middleware/	TC-SEC-01	ST-SEC-1
42	SEC-2	All client-server traffic shall use HTTPS with TLS 1.2 or higher.	SAD 3.9 / 3.7	SAD 4.4	server/src/middleware/	TC-SEC-02	ST-SEC-2
43	SEC-3	A resume file shall be served only to its owner, to recruiters of...	SAD 3.9 / 3.7	SAD 4.4	server/src/middleware/	TC-AUTH-04	ST-SEC-3
44	SEC-4	All request bodies shall be validated against a schema, and query...	SAD 3.9 / 3.7	SAD 4.4	server/src/middleware/	TC-SEC-01	ST-SEC-4
45	SEC-5	Login and registration endpoints shall be rate limited to 10...	SAD 3.9 / 3.7	SAD 4.4	server/src/middleware/	TC-SEC-02	ST-SEC-5

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
46	SEC-6	Every application status change and every admin action on accounts...	SAD 3.9 / 3.7	SAD 4.4	server/src/middleware/	Review	ST-SEC-6